Department of Computer Science Engineering

Chitkara University

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Memories-App

Submitted By

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Statement of Submission

This is to certify that **Aditya Sood**, (Roll Number 2010991786), has successfully completed the final project, for the subject 'Front End Engineering', named as '**Memories App**', at the Department of Computer Science Engineering, Chitkara University, to fulfil the requirement of the degree 'Bachelor of Engineering (Computer Science)'.

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Acknowledgement

It gives me a great sense of pleasure to present the report of the project work undertaken during the third year of my degree. I owe special debt of gratitude to my Project Coordinator – Mr. Rafeeq Qureshi, Chitkara University, Punjab for his constant support and guidance throughout the course of my work. It is only his cognizant efforts that my endeavours have seen light of the day.

I also take the opportunity to acknowledge the contribution of Dr. Susheela Hooda, Course Coordinator – Front End Engineering, Chitkara University, Punjab for her full support and assistance during the development of this project.

I would not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of my project.

Finally, I would like to convey my sincere thanks to my batchmates; without them, the task would not have been accomplished in such a timely manner.

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Abstract

At its core, social media holds out the promise of connection. People use social media to stay in touch with their family and friends; and reconnect with their old friends. The proposed project is a social media app, 'Memories App'. This paper illustrates the features, development method, result and uses of this application. The key concept behind this project is that it allows its users to share interesting events that take place in their lives. The user can add a new post by using the option of 'creating a memory' where he/she must add the creator's name, the title of the post, any message that needs to be added, and finally, the tags that would go along with the image. After completing the necessary steps, the post becomes visible on the user's database. The app also allows the users to edit the post after it is posted. This can be done by clicking on the three dots on the post. Other features include liking the post and deleting a post.

This app would allow its users to stay in touch with someone they cannot be physically together with. They can look through each other's memories and reminisce. It's not necessary that everyone from a family or a close group of friends stay together, hence, it can be very difficult for everyone to get involved in each other's lives. This platform lets you share as many memories as you want and let your close ones feel that they are equally involved in your life.

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Introduction

Memories App is a social media application where the user can share the happenings of his/her life by uploading a picture of the same. The application is easy to use and user friendly; in the sense that the user must first sign up on the application to create an account. After that, he/she can simply add new posts to their feed by following the below steps –

1. Write the name of the creator of the post, i.e., their name.
2. Add a title to the post. For example – if the picture is of the user's office, he/she can title the post as 'A Day at Work' or 'Daily Scenes of Office', depending on his/her preference.
3. The user may add a message to the post also, which can be a brief description of the picture.
4. Next step is to add relevant hashtags to the post.
5. The main part comes next, adding the image.
6. Users can post the image to their feed by clicking on the option 'Submit'.

Post performing the above steps, the post will be visible in the database. The list of options available to the user doesn't end here. He/she can also edit the post if necessary and can also delete it.

Project Aim and Objectives

The main aim of 'Memories App' is to provide a space to the users where they can share the bits and parts of their lives and stay connected with their family and friends. It also aims to be a platform for people where they feel they are involved in their family members' or friends' lives – especially in this busy world where people generally don't have time to get in touch with everyone, everyday on a call.

Objectives of Memories App –

1. Share media content online with friends
2. Connect family and friends
3. Create a personal space for the user where he/she feels comfortable to share the pictures
4. Update the user's friends on the app about his/her life
5. Provide a platform where the users of the app can start an online business
6. Users can search their near and dear ones and send them a request to connect with them

User Requirements

This application can be used by anyone who is interested to share glimpses of his/her life by posting images on an app.

By using the Memories App, the operation of involving people in your lives is very easy and requires a lot less effort. This system provides a user-friendly application with a 'create' tab which contains all the things that the user must provide to make a post. It makes the input entry way easier to understand and use. It is also created to ensure that the posts that a user shares are only visible to the friends that he/she has added to his/her list to ensure security.

This system will store all the posts that the users have made in its database and the users can revisit them anytime they want. It also provides search option to help users find their family or friends who they can add to their accounts by sending them a request.

The users also have the option to edit the post in case something is wrong with it. Therefore, they can change the message associated with the post, add, or remove hashtags, change the title of the post etc. Another option available to the users is to delete the post altogether.

Background of Project

We are dwelling in an era, in an age, where it is possible to be connected and to stay in touch with every one of our acquaintances, idols, our near and dear ones. Even though we may not be physically present to communicate, we can still be connected to everyone we know – our childhood pals, high school friends, college batchmates, teachers, our loved ones staying abroad – the list goes on.

Also, through social media, a single post can get viral and reach hundreds and thousands of people at the same time through their phones. A single video, having the right kind of content, can easily cross a million views. In the 21st century, social media plays an important role in defining the success or failure of a business.

Be it a small business or a big one, they all see quite a profit potential through social media. This is only possible due to its easy availability, accessibility, and response due to the millions of people utilizing social media websites and apps for various purposes.

This is where the Memories App also comes in. The app ensures that all the things discussed above, and the requirements are met. Therefore, with this computerised system, sharing memories has become very easy for people. They can stay connected to their loved ones as much as they want. This app will also help them to get a wider reach and a higher number of customers for their businesses with a single post, which would have been difficult otherwise by just advertising the same through offline channels of marketing.

System Analysis

Here, we will discuss and analyse about the developing process of the Memories App including software requirement specification and comparison between the existing and proposed system. The functional and non-functional requirements are included in the software requirement specifications to provide complete description and overview of the system requirement before the developing process is carried out. Besides that, existing vs proposed provides a view of how the proposed system will be more efficient than the existing one.

A. Software Requirement Specification

1. **Product description –**
   Memories App is a computerized system which helps the user to share videos and photos on an app to stay connected with friends or family. It reduces the difficulty of staying touch with other people.

2. **Problem statement –**
   The problems that tended to occur before having a system where photos can be share for updating friends/family on one's whereabouts include –
   - Having to call people every other day if you want to stay in touch
   - Not being able to show anyone visually what is happening in your life
   - Having to put efforts in offline advertising to update people of your business
   - Sharing the product or service images and information through other channels which prove to be more costly and time taking

3. **Proposed system objectives**
   If the proposed system is implemented, the following things would take place that would prove to be a solution to the above stated problems –
   - Only a picture a day can be uploaded to let your friends/family know what you're up to
   - Instead of telling people what happened with you, you can just share a snap of that on the app
   - People can advertise their businesses on the application for a wider reach
   - It will save cost since with a few clicks on the mouse, and few search keywords, a user can share pictures of the product/service offered and let people know about it

System Requirements

1. **Non-Functional Requirements**
   - Efficiency Requirement – When this system will be implemented, the user can easily post and access the posts of other people, since searching for the people and getting in touch with them will be much faster.
   - Reliability Requirement – The system should accurately perform user registration and sharing posts.
   - Usability Requirement – The system is designed for a user-friendly environment so that people can perform the various tasks easily and in an effective way.

2. **Functional Requirements**
   - User Login – This feature is used by the user to login into the system. They are required to enter the user ID and password before they are allowed to enter the system. The user ID and password will be verified and if they are invalid, the user will not be allowed to enter the system. The credentials will be provided when the user registers in the system. The system must only allow users with valid credentials to enter the system.
   - Register New User – This feature can be performed by all users to register new users to create accounts. The system must be able to verify the information provided and delete it if the information provided is wrong.
   - Add a New Post – The user can add a new post into the database by clicking on the button that says 'create' and then give details like name of the creator, title, message, tags, image etc.
   - Edit or Delete a Post – Once a post is made, the user can also edit or delete it if he/she is not satisfied with how it turned out.

Technologies Used

1. **Front-end (client side)**
    - React
        - React-Router-Dom
        - Axios
        - Material – UI/Core
        - Redux

2. **Back-end (server side)**
    - MongoDB
        - Mongoose
    - Express
    - NodeJS
    - Bodyparser
    - Nodemon

3. **Common**
    - JsonWebToken

Schemas

1. Post Schema

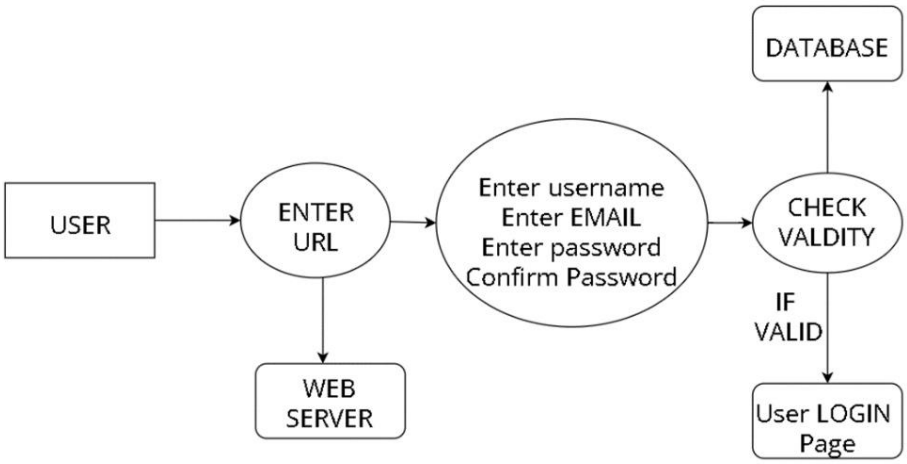
```
const postSchema = mongoose.Schema({
    title: String,
    message: String,
    name: String,
    creator: String,
    tags: [String],
    selectedFile: String,
    likes: { type: [String], default: [] },
    createdAt: {
        type: Date,
        default: new Date(),
    },
})
```

2. After Schema

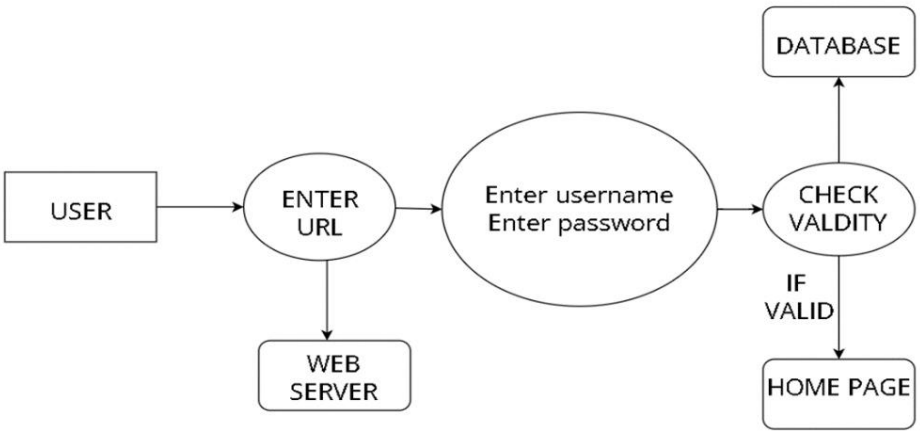
```
const userSchema = mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true },
  password: { type: String, required: true },
  id: { type: String },
});
```

Data Flow Diagrams

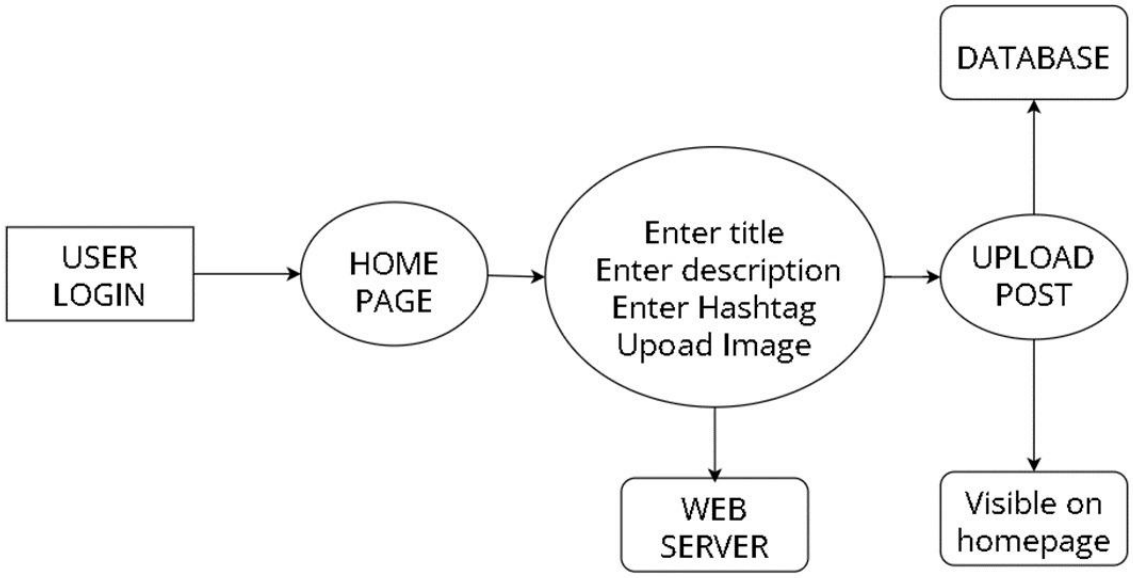
Sign-Up



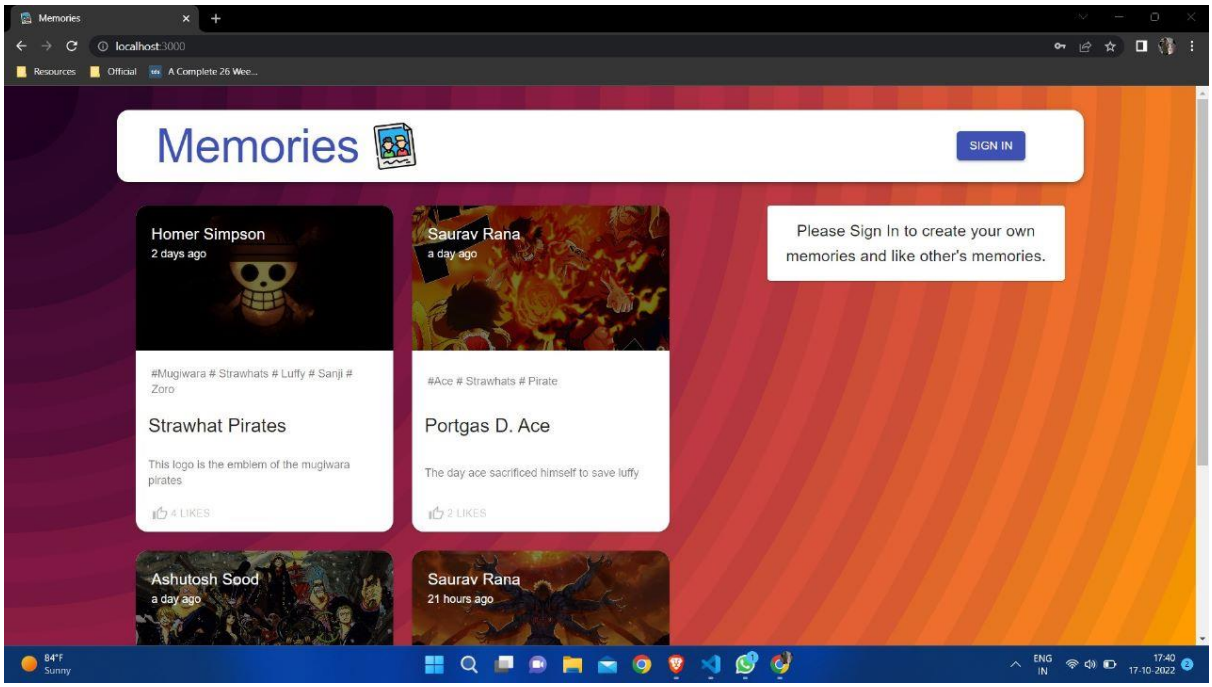
Sign-In



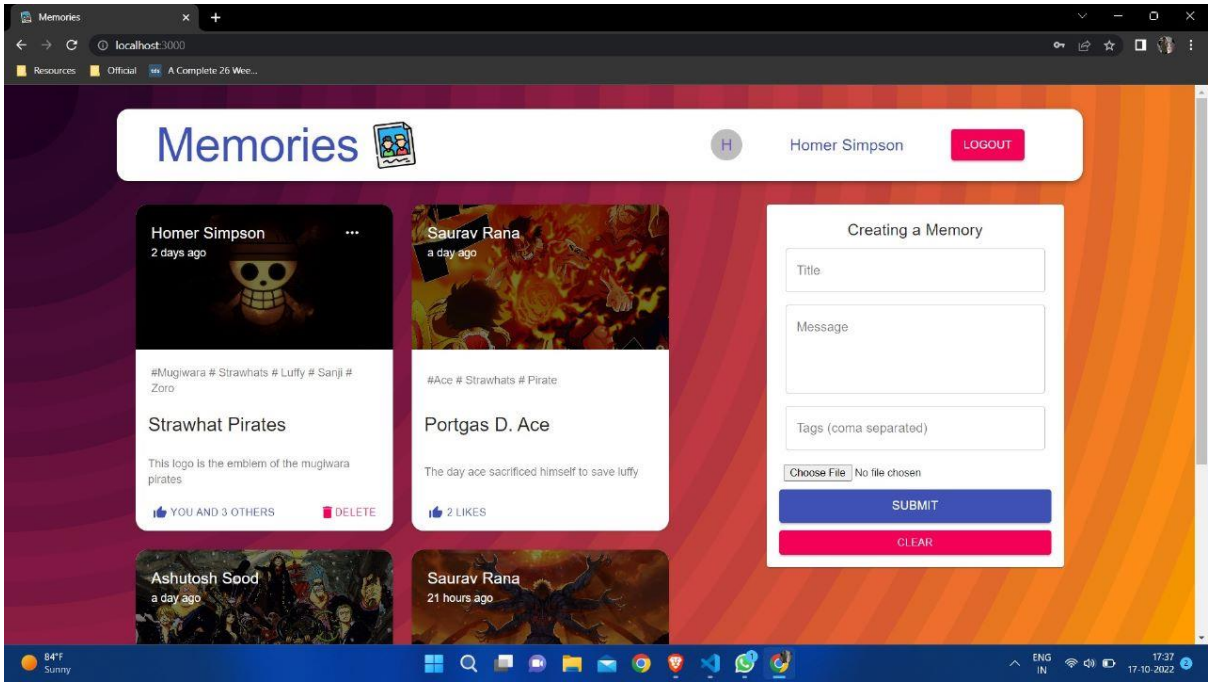
Create Post


DATABASE
USER LOGIN
HOME PAGE
Enter title
Enter description
Enter Hashtag
Upoad Image
UPLOAD POST
WEB SERVER
Visible on homepage

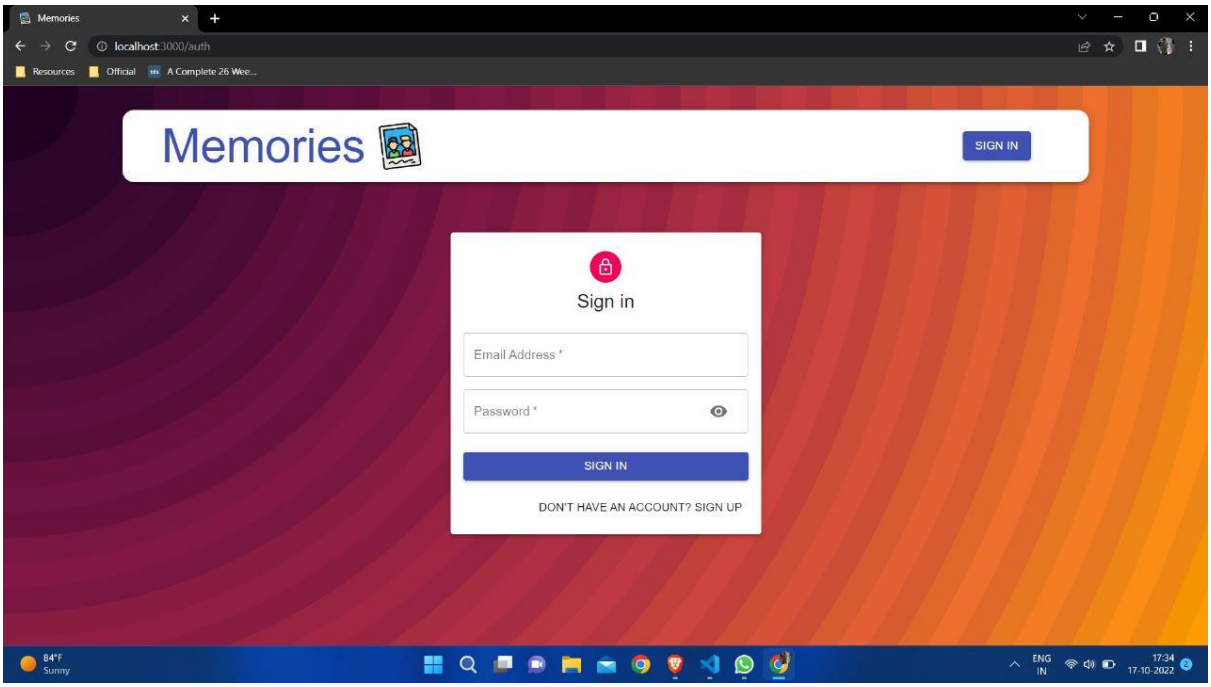
Landing Page



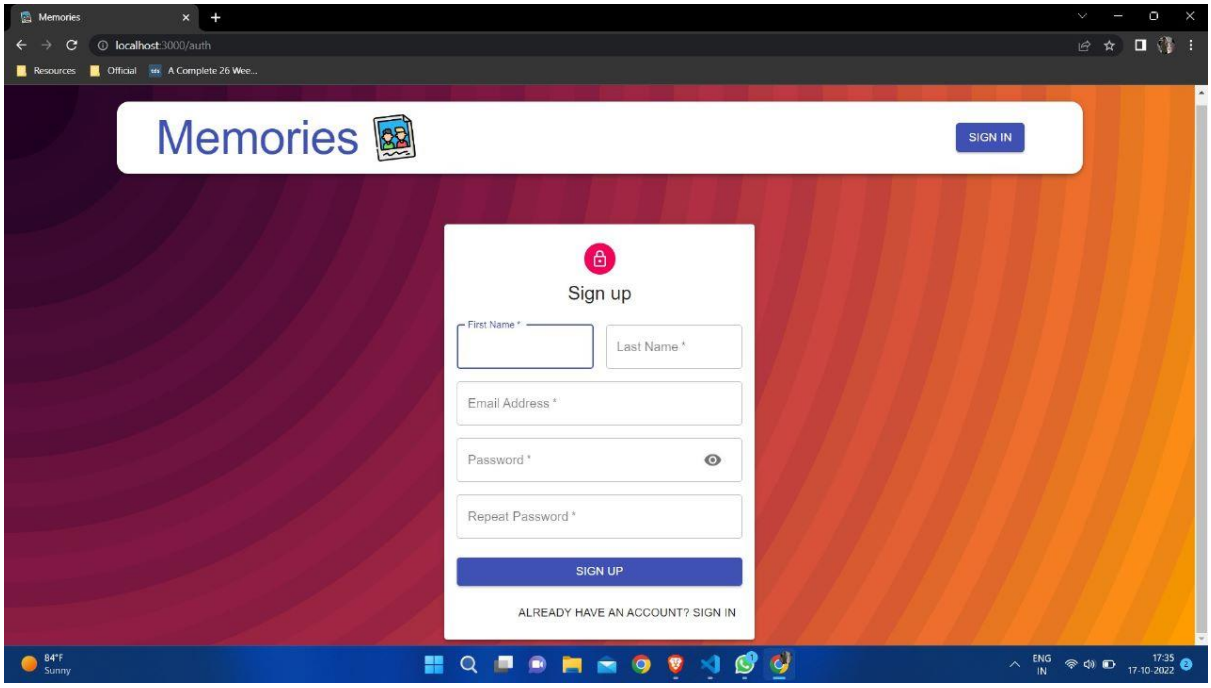
Home Page



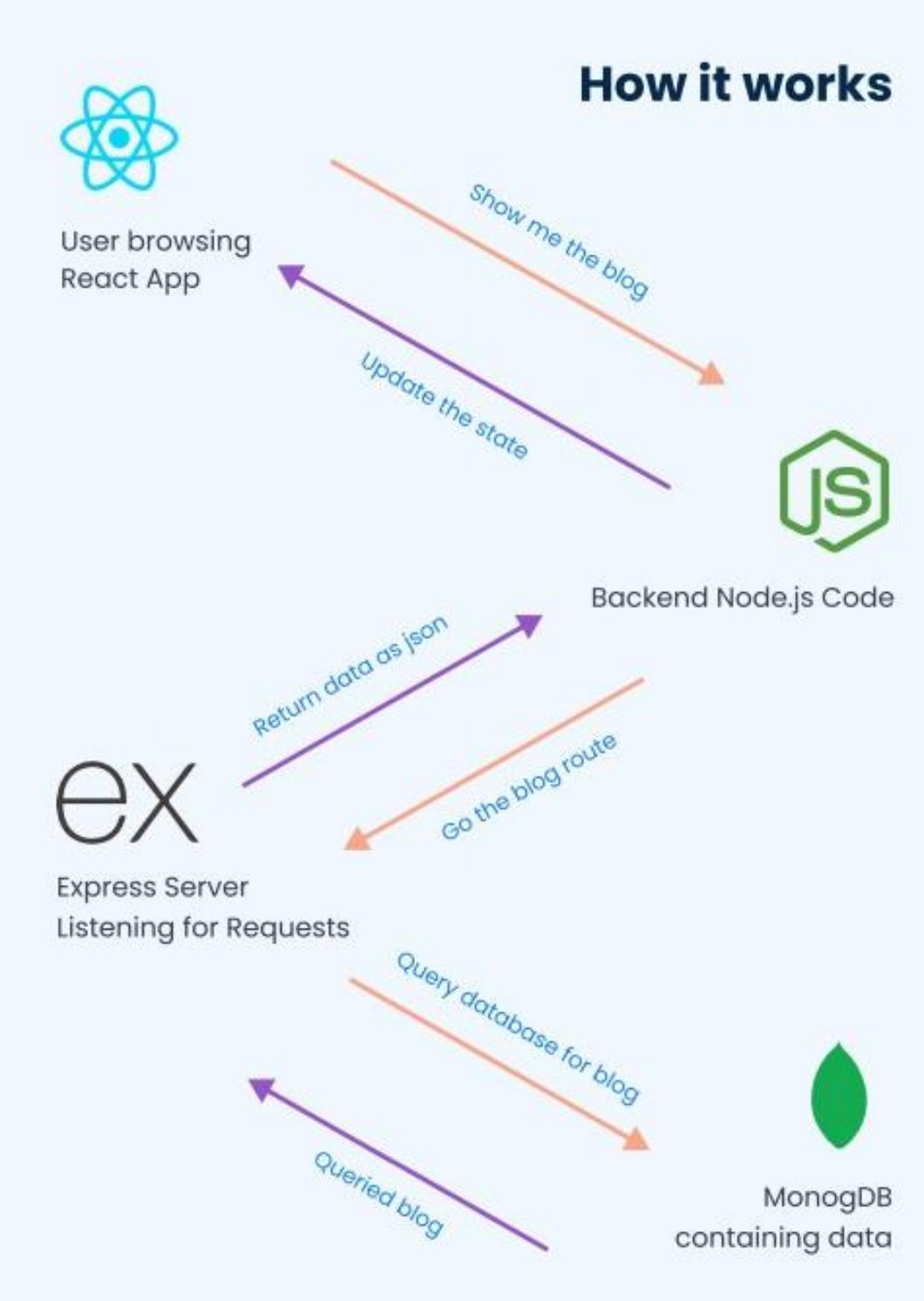
Sign in Page



Sign-up Page



Workflow of the App



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